Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 1 / 方法 1 [AL03-M01]

Choose coordinates for addition, polar form for multiplication, and geometric loci for a modulus or argument condition. The useful bridge is the complex quotient: its modulus is a length ratio and its argument is a directed angle.

加法用直角坐标，乘法用极形式，模长或辐角条件用几何轨迹。关键桥梁是复商：其模长为长度比，其辐角为有向角。

Separate real and imaginary parts with all sign branches

连同符号分支分离实部与虚部

Procedure

1. Write $z = x + iy$, expand using $i^2 = -1$, and compare real and imaginary parts.

2. Before clearing a complex denominator, exclude its zero. For principal square roots of real quantities, split nonnegative and negative radicands.

3. For product systems, introduce the common product and reconstruct both consistent square-root signs.

4. Check modulus, positivity or integrality restrictions on the resulting coordinates.

通用步骤

1. 写 $z = x + iy$，用 $i^2 = -1$ 展开，再比较实部与虚部。

2. 清除复分母前先排除零；对实数量的主平方根，区分被开方式非负与负。

3. 乘积方程组可设置共同乘积，并恢复两个一致的平方根符号。

4. 检查所得坐标的模长、正性或整数限制。

> Example. Find all complex z with $z + 2\overline{z} = 6 + i$.

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 1 / 方法 1 [AL03-M01] (continued / 续)

> 例题：求满足 $z+2\overline{z}=6+i$ 的全部复数 z。

Worked solution

1. Write $z=x+iy$, so $\overline{z}=x-iy$.

2. The left side is $3x-iy$. Comparing components gives $3x=6, -y=1$.

3. The unique solution is $z=2-i$, and direct substitution gives $6+i$.

例题解析

1. 写 $z=x+iy$，则 $\overline{z}=x-iy$。

2. 左侧为 $3x-iy$，比较分量得 $3x=6, -y=1$。

3. 唯一解为 $z=2-i$，直接代入得 $6+i$。

A complex number is real exactly when its imaginary part vanishes, including the value zero.

复数为实数当且仅当虚部为零，其中包括数值零。

For a Gaussian integer $a+bi$, the norm is a^2+b^2, and its cube has components a^3-3ab^2 and $3a^2b-b^3$. Multiplication by a unit $\pm1, \pm i$ and conjugation change signs or exchange component magnitudes. Before a triangle test, verify both candidate components are positive and use $|X-Y|<L<X+Y$ with proved finite bounds on a, b.

高斯整数 $a+bi$ 的范数为 a^2+b^2，立方的分量为 $a^3-3ab^2, 3a^2b-b^3$。乘单位 $\pm1, \pm i$ 与共轭可改变符号或交换分量绝对值。检验三角形前，须确认候选分量为正，并使用 $|X-Y|<L<X+Y$，且先证明 a, b 的有限界。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 1 / 方法 1 [AL03-M01] (continued / 续)

Practice / 配套练习: AIME-2007-I-Q03 (WB p. 62), AIME-2009-I-Q02 (WB p. 63), AIME-2012-II-Q08 (WB p. 75), AIME-2016-I-Q07 (WB p. 79), AIME-2018-II-Q05 (WB p. 64), AIME-2019-II-Q08 (WB p. 82), AIME-2023-I-Q15 (WB p. 96)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 2 / 方法 2 [AL03-M02]

Pair conjugate roots and recover their real polynomial factor

配对共轭根并恢复实系数因子

Procedure

1. A real-coefficient polynomial contains every nonreal root together with its conjugate and equal multiplicity.

2. The pair $u \pm iv$ gives factor $x^2 - 2ux + (u^2 + v^2)$.

3. Use Vieta to connect that factor with the remaining roots. For an integer polynomial, trace and norm of this pair are integral only when the pair is known to form a monic rational quadratic factor; Gauss's lemma then applies.

4. Separate all-real multisets from the one-nonreal-pair case when counting cubics.

通用步骤

1. 实系数多项式的每个非实根都与其共轭以相同重数出现。

2. 根对 $u \pm iv$ 给出因子 $x^2 - 2ux + (u^2 + v^2)$。

3. 用韦达将该因子与其余根联系。整系数多项式中，只有已知该根对构成首一有理二次因子时，才能由高斯引理要求其迹与范数为整数。

4. 统计三次式时，将全实根多重集与一个非实根对的情形分开。

> Example. A monic real cubic has roots $2 + i$ and -3. Its x^2 coefficient is -1. Find the constant term.

> 例题：首一实三次式有根 $2 + i, -3$，其 x^2 系数为 -1，求常数项。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 2 / 方法 2 [AL03-M02] (continued / 续)

Worked solution

1. The third root must be $2-i$. The root sum is $2+i+2-i-3=1$, consistent with coefficient -1.

2. The conjugate pair has product $(2+i)(2-i)=5$.

3. The three-root product is -15, so the constant term is its negative, 15.

例题解析

1. 第三根必为 $2-i$。根和为 $2+i+2-i-3=1$，与系数 -1 一致。

2. 共轭根对乘积为 $(2+i)(2-i)=5$。

3. 三根乘积为 -15，所以常数项为其相反数 15。

Conjugation fixes the polynomial only when its coefficients are real.

只有系数为实数时，共轭才保持该多项式不变。

Integer coefficients alone do not force each conjugate pair to have integral trace or norm: in x^3-2, the nonreal pair has irrational sum $-\sqrt[3]{2}$. Apply coefficient integrality only after establishing a rational factor, or use the full polynomial coefficient equations.

整系数并不自动使每个共轭根对的迹与范数为整数。例如 x^3-2 的非实根对之和为无理数 $-\sqrt[3]{2}$。须先确定因子系数为有理数，才能使用因子整系数性；否则应使用整个多项式的系数方程。

Practice / 配套练习: AIME-2005-I-Q06 (WB p. 68), AIME-2010-II-Q07 (WB p. 71), AIME-2013-I-Q10 (WB p. 76), AIME-2013-II-Q12 (WB p. 56), AIME-2021-II-Q04 (WB p. 36), AIME-2026-II-Q11 (WB p. 98)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 3 / 方法 3 [AL03-M03]

Use polar powers and unit-circle angle congruences

使用极形式幂与单位圆辐角同余

Procedure

1. For nonzero z, write $z = re^{i\theta}$; multiplication adds arguments and multiplies moduli.

2. Equality of unit complex numbers is an angle congruence modulo 2π, not an ordinary angle equality.

3. For reality, set the imaginary part to zero or equate a complex expression to its conjugate.

4. Count a half-open complete argument interval and remove overlapping angle families explicitly.

通用步骤

1. 非零 z 写成 $re^{i\theta}$；相乘时辐角相加、模长相乘。

2. 单位复数相等对应模 2π 的辐角同余，而非普通角度相等。

3. 要求实数时，可令虚部为零，或令表达式与其共轭相等。

4. 在半开完整辐角区间计数，并明确消除不同辐角族的重叠。

> Example. How many unit complex numbers satisfy $z^4 \in \mathbb{R}$ but $z^2 \notin \mathbb{R}$?

> 例题：有多少单位复数满足 $z^4 \in \mathbb{R}$ 而 $z^2 \notin \mathbb{R}$?

Worked solution

1. Write $z = e^{i\theta}$ for $0 \le \theta < 2\pi$. Reality of z^4 requires 4θ a multiple of π, giving eight angles.

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 3 / 方法 3 [AL03-M03] (continued / 续)

2. Among them z^2 is real exactly when 2θ is a multiple of π, giving four of those angles.

3. The required difference is $8 - 4 = 4$.

例题解析

1. 写 $z = e^{i\theta}$，其中 $0 \le \theta < 2\pi$。z^4 为实数要求 4θ 为 π 的倍数，给出八角。

2. 其中 z^2 为实数恰当 2θ 为 π 的倍数，共四角。

3. 所求个数为 $8 - 4 = 4$。

A root may have modulus one without being a primitive root of its stated order.

根的模为一，并不表示它一定是该次数的本原单位根。

Practice / 配套练习: AIME-2000-II-Q09 (WB p. 67), AIME-2001-II-Q14 (WB p. 86), AIME-2005-II-Q09 (WB p. 69), AIME-2012-I-Q06 (WB p. 73), AIME-2014-II-Q10 (WB p. 78), AIME-2018-I-Q06 (WB p. 81), AIME-2019-II-Q08 (WB p. 82), AIME-2022-I-Q04 (WB p. 65)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 4 / 方法 4 [AL03-M04]

Sum rotating displacements with a complex geometric series

以复等比级数累加旋转位移

Procedure

1. Encode the multiplier for one displacement as $q = re^{i\theta}$.

2. For finite motion with $q \neq 1$, use $\sum_{j=0}^{n-1} q^j = (1-q^n)/(1-q)$; if $q = 1$, the sum is n. For a limit verify $|q| < 1$ first.

3. A rotation about c is $z \mapsto c + e^{i\theta}(z-c)$; composing with translations produces an affine map.

4. Compute the final norm only after summing the vectors; for nested disks prove the limit lies in every disk.

通用步骤

1. 将一次位移的倍率编码为 $q = re^{i\theta}$。

2. 有限运动中，$q \neq 1$ 时用 $\sum_{j=0}^{n-1} q^j = (1-q^n)/(1-q)$；$q = 1$ 时和为 n。极限问题先验证 $|q| < 1$。

3. 绕 c 旋转为 $z \mapsto c + e^{i\theta}(z-c)$；与平移复合得到仿射映射。

4. 向量求和后才求最终模长；嵌套圆盘还须证明极限位于每个圆盘内。

> Example. Find the fixed point of $F(z) = iz + 1$ and explain the geometry of all its orbits.

> 例题：求 $F(z) = iz + 1$ 的不动点，并说明全部轨道的几何意义。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 4 / 方法 4 [AL03-M04] (continued / 续)

Worked solution

1. The fixed point satisfies $c = ic + 1$, so $c = 1/(1-i) = (1+i)/2$.

2. Subtraction gives $F(z) - c = i(z - c)$, a counterclockwise quarter-turn about c.

3. The fixed point has period one. Every other point has exact period four, because $i^k(z - c) = z - c$ first holds at $k = 4$.

例题解析

1. 不动点满足 $c = ic + 1$，故 $c = 1/(1-i) = (1+i)/2$。

2. 相减得 $F(z) - c = i(z - c)$，即绕 c 逆时针四分之一周旋转。

3. 不动点周期为一；其余每点的准确周期为四，因为 $i^k(z - c) = z - c$ 首次在 $k = 4$ 成立。

Adding lengths gives path length; adding complex displacements gives the endpoint.

长度相加得到路程，复位移相加才得到终点。

If nested disks have radii tending to zero and centers converging to B, verify $|B - O_n| \le r_n$ for every n. Any two common points would have distance at most $2r_n \to 0$, proving uniqueness.

嵌套圆盘半径趋零、圆心趋于 B 时，须对每个 n 验证 $|B - O_n| \le r_n$。任意两个公共点距离至多为 $2r_n \to 0$，故公共点唯一。

Practice / 配套练习: AIME-2008-II-Q09 (WB p. 70), AIME-2013-I-Q14 (WB p. 93), AIME-2017-II-Q12 (WB p. 94), AIME-2020-I-Q08 (WB p. 83), AIME-2026-I-Q05 (WB p. 66)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 5 / 方法 5 [AL03-M05]

Evaluate products over roots of unity

计算单位根上的乘积

Procedure

1. Use $x^n-1=\prod_{k=0}^{n-1}(x-\omega^k)$ when the powers exhaust all nth roots.

2. For nontrivial roots, divide out $x-1$ and evaluate its removable value at one as n.

3. Factor a general polynomial into its roots, interchange the two products and reduce powers modulo the smaller defining polynomial.

4. For chord lengths, take the modulus after the complex product; do not replace a signed algebraic product by a length prematurely.

通用步骤

1. 当各幂穷尽所有 n 次单位根时，使用 $x^n-1=\prod_{k=0}^{n-1}(x-\omega^k)$。

2. 对非平凡单位根，先除去 $x-1$，在一处的可去值为 n。

3. 一般多项式先按其根分解，交换两重乘积，再以较小的定义多项式约化幂。

4. 弦长问题应在复乘积之后取模，不能过早把带符号代数乘积换成长度。

> Example. For a primitive fifth root ω, compute $\prod_{k=1}^{4}(1-\omega^k)$ and the product of distances from 1 to those four roots.

> 例题：对本原五次单位根 ω，计算 $\prod_{k=1}^{4}(1-\omega^k)$，及一到这四根的距离之积。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 5 / 方法 5 [AL03-M05] (continued / 续)

Worked solution

1. The polynomial of the nontrivial fifth roots is $1+x+x^2+x^3+x^4$.

2. Evaluating at one gives the algebraic product 5.

3. The distance product is the modulus of that product, also 5.

例题解析

1. 非平凡五次单位根的多项式为 $1+x+x^2+x^3+x^4$。

2. 在一处代值得代数乘积 5。

3. 距离乘积为该乘积的模长，也为 5。

If the chosen root is not primitive, its powers may repeat and miss roots.

若所选根非本原根，其幂可能重复并遗漏其他根。

Practice / 配套练习: AIME-2003-II-Q15 (WB p. 88), AIME-2004-I-Q13 (WB p. 89), AIME-2009-II-Q13 (WB p. 91), AIME-2023-II-Q08 (WB p. 84), AIME-2024-II-Q13 (WB p. 97)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 6 / 方法 6 [AL03-M06]

Translate complex quotients into angles, circles and perpendicularity

将复商转为角、圆与垂直条件

Procedure

1. For nonzero displacements u, v, u/v real means parallel directions; purely imaginary means perpendicular directions.

2. A real cross-ratio equates directed inscribed angles modulo π; for three fixed noncollinear points its locus is their circle.

3. Equal-distance loci are perpendicular bisectors; a modulus equation $|z-a|=R$ is a circle.

4. Exclude quotient poles and zero displacement legs, then use coordinates or a circle center for the requested extremum.

通用步骤

1. 对非零位移 u, v，u/v 为实数表示方向平行，为纯虚数表示方向垂直。

2. 实交比表示有向圆周角模 π 相等；三个固定点不共线时，轨迹为其外接圆。

3. 等距轨迹是垂直平分线，模方程 $|z-a|=R$ 是圆。

4. 排除商的极点与零位移边，再用坐标或圆心求所需最值。

> Example. Find the locus of z for which $(z-1)/(z+1)$ is purely imaginary, including zero as an imaginary value.

> 例题：求使 $(z-1)/(z+1)$ 为纯虚数的 z 的轨迹，其中允许纯虚数取零。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 6 / 方法 6 [AL03-M06] (continued / 续)

Worked solution

1. The pole $z=-1$ is excluded. Write $z=x+iy$ and multiply numerator and denominator by $x+1-iy$.

2. The real numerator is x^2+y^2-1 and the denominator is positive, so reality of the zero real part requires $x^2+y^2=1$.

3. The locus is the unit circle with -1 deleted; $z=1$ remains because the quotient is zero.

例题解析

1. 排除极点 $z=-1$。写 $z=x+iy$，分子分母乘以 $x+1-iy$。

2. 分子实部为 x^2+y^2-1，分母为正，故实部为零要求 $x^2+y^2=1$。

3. 轨迹为删去 -1 的单位圆；$z=1$ 保留，因为商为零。

Specify whether zero counts as purely imaginary before using it as a perpendicularity test.

用纯虚数判垂直之前，须明确零是否允许；非退化垂直位移不能为零。

Practice / 配套练习: AIME-2012-I-Q14 (WB p. 92), AIME-2017-I-Q10 (WB p. 80), AIME-2019-I-Q12 (WB p. 95), AIME-2025-I-Q08 (WB p. 382)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 7 / 方法 7 [AL03-M07]

Invert a half-plane and carry unions through the map

反演半平面，并把并集随映射传递

Procedure

1. For $w = 1/z = u + iv$, substitute $z = (u - iv)/(u^2 + v^2)$ with $w \neq 0$.

2. Multiply an inequality only by the positive denominator $u^2 + v^2$, then complete the square.

3. Preserve strictness and boundary exclusions. The image of a union is the union of the images.

4. For a union of symmetric disks, identify which disk reaches farthest on each ray before computing wedge areas.

通用步骤

1. 对 $w = 1/z = u + iv$，代入 $z = (u - iv)/(u^2 + v^2)$，其中 $w \neq 0$。

2. 只以正分母 $u^2 + v^2$ 乘不等式，再配方。

3. 保留严格性与边界排除；并集的像是各像之并。

4. 对对称圆盘并，先确定各射线方向上哪个圆盘伸得最远，再求扇形分区面积。

> Example. Find the image of the half-plane $\Re z > 1$ under $w = 1/z$.

> 例题：求半平面 $\Re z > 1$ 在 $w = 1/z$ 下的像。

Worked solution

1. The inequality becomes $u/(u^2 + v^2) > 1$, with a positive denominator.

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 7 / 方法 7 [AL03-M07] (continued / 续)

2. Rearrange to $(u-1/2)^2+v^2<1/4$.

3. The image is the open disk centered at $1/2$ of radius $1/2$. Its boundary includes zero, so zero is already excluded by strictness.

例题解析

1. 不等式化为 $u/(u^2+v^2)>1$，分母为正。

2. 整理得 $(u-1/2)^2+v^2<1/4$。

3. 像为以 $1/2$ 为圆心、半径 $1/2$ 的开圆盘；零位于其边界，已由严格性排除。

> The map 1/z includes conjugation relative to ordinary radial inversion; use the actual algebra.

> 映射 1/z 相对通常径向反演还含共轭，应使用实际代数表达式。

Practice / 配套练习: AIME-2008-II-Q13 (WB p. 90)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 8 / 方法 8 [AL03-M08]

Optimize a complex norm or real part and prove equality

优化复模或实部，并证明等号可达

Procedure

1. Factor out a fixed modulus or write $z = R(\cos\theta + i\sin\theta)$.

2. For distance from a fixed point to a circle, align the radius oppositely for a maximum; for a real part use $A\cos\theta + B\sin\theta \le \sqrt{A^2+B^2}$.

3. For independent choices, optimize each summand separately. For an argument range, use tangent rays to the locus.

4. Check the equality point respects half-plane, sign and excluded-point restrictions.

通用步骤

1. 提出固定模长，或写 $z = R(\cos\theta + i\sin\theta)$。

2. 固定点到圆的最远距离使半径方向相反；实部最值用 $A\cos\theta + B\sin\theta \le \sqrt{A^2+B^2}$。

3. 独立选择时分别优化每一项；辐角范围则用轨迹的切射线。

4. 检查等号点符合半平面、符号与排除点限制。

> Example. For $|z| = 1$, find the maximum of $\Re((3+4i)z)$ and a maximizing z.

> 例题：当 $|z| = 1$，求 $\Re((3+4i)z)$ 的最大值及一个达到值的 z。

Worked solution

1. With $z = x + iy$, the objective is $3x - 4y$ and $x^2 + y^2 = 1$.

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 8 / 方法 8 [AL03-M08] (continued / 续)

2. Cauchy bounds it by $\sqrt{3^2+(-4)^2}=5$.

3. Take $z=(3-4i)/5$ to attain five, with modulus one as required.

例题解析

1. 设 $z=x+iy$，目标为 $3x-4y$，且 $x^2+y^2=1$。

2. 柯西不等式给出上界 $\sqrt{3^2+(-4)^2}=5$。

3. 取 $z=(3-4i)/5$ 达到五，其模长确为一。

A norm bound needs the direction of equality, not just the numerical upper bound.

模长估界须给出等号方向，而不只是数值上界。

Practice / 配套练习: AIME-2011-II-Q08 (WB p. 72), AIME-2012-II-Q06 (WB p. 74), AIME-2014-I-Q07 (WB p. 77), AIME-2024-I-Q07 (WB p. 85)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 9 / 方法 9 [AL03-M09]

Compose fractional maps and control finite polynomial-root orbits

复合分式映射，并控制有限多项式根轨道

Procedure

1. Compose the exact map symbolically, keeping every pole and any excluded intermediate value.

2. If a small iterate is identity, reduce the index only for orbits proved to avoid the poles.

3. For a polynomial root map, a finite invariant root set cannot contain a point whose modulus grows without bound.

4. Combine root-product modulus and conjugation to restrict the final root set, then substitute back into the original identity.

通用步骤

1. 符号复合精确映射，并保留每个极点及被排除中间值。

2. 若某个低次迭代为恒等，只能对已证明避开极点的轨道约化下标。

3. 多项式根映射的有限不变根集不能含模长无限增长的点。

4. 结合根乘积模长与共轭限制最终根集，再代回原恒等式。

> Example. For $F(z)=1/(1-z)$, show the orbit of $z_0=2$ has period three.

> 例题：对 $F(z)=1/(1-z)$，证明 $z_0=2$ 的轨道周期为三。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 9 / 方法 9 [AL03-M09] (continued / 续)

Worked solution

1. The successive values are $2, -1, 1/2, 2$.

2. None equals the pole one, so all intermediate maps are defined.

3. The first three values are distinct and the fourth returns to the first, proving exact period three.

例题解析

1. 逐次值为 $2, -1, 1/2, 2$。

2. 各值均不为极点一，因此所有中间映射都有定义。

3. 前三个值互异，第四个返回首值，故准确周期为三。

An algebraic composition formula may cancel a pole that the original step still cannot cross.

复合公式可能约掉某极点，但原逐步映射仍不能经过该点。

Practice / 配套练习: AIME-2002-I-Q12 (WB p. 87), AIME-2007-II-Q14 (WB p. 211)